INCH-POUND

MIL-PRF-51560A(EA)
1 July 1997
SUPERSEDING
MIL-C-51560
6 June 1994

PERFORMANCE SPECIFICATION

CANISTER, CHEMICAL-BIOLOGICAL MASK: C2A1

This specification is approved for use by the U.S. Army Edgewood Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of mask canister used to protect against chemical agents, biological agents, and radioactive dust.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.1.1 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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U.S. ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-1-1515	-	Cover, Canister Carrier
5-1-1517	-	Shell, Canister Carrier
5-1-1960	-	Disconnect, Armor Quick Assembly
26-19-51	-	Machine, Rough Handling, Q113
38-6-351-8	-	Indicator, Canister Resistance, Q112
125-8-1-2	-	Indicator, Water Repellency, Q101
136-41-600-1	-	All Purpose Apparatus, Gas Life Testing, Q95
136-41-3500	-	Fifty Liter Per Minute Filter Life Tester, Q250
136-42-102	-	Penetrometer, Filter Testing, Q127
136-45-842	-	Tester, Canister Shaker Assembly, Q261
136-301-38	-	Tester, Air Leakage (Dry Bubble), Q204

PURCHASE DESCRIPTIONS

EA-C-1704	-	Carbon, Activated, Impregnated, Triethylenediamine-Copper-Silver-Zinc-Molybdenum (TEDA-ASZM)

(Copies are available from Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-M, Aberdeen Proving Ground, MD 21010-5423.)

CODE OF FEDERAL REGULATIONS

40 CFR Part 261	-	Hazardous Waste Management

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

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AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS

ISBN-0-936712-546 - Threshold Limit Values for Chemical Substances and Physical Agents in the Work Environment and Biological Exposure Indices

(Application for copies should be addressed to American Conference of Governmental Industrial Hygienists, Building D-7, 6500 Glenway Avenue, Cincinnati, OH 45211.)

ASTM STANDARDS

B 117 - Salt Spray (Fog) Testing
D 2020 - Mildew (Fungus) Resistance of Paper and Paperboard
D 2867 - Moisture in Activated Carbon

(Application for copies should be addressed to ASTM, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials and components. The canister shall be constructed so that it conforms to the requirements of this specification. All materials used shall be non-magnetic. The contractor shall select materials for use in the canister which pose no potential inhalation or skin contact hazard to the user in excess of the limits as recommended by the American Conference of Governmental Industrial Hygienists (ISBN 0-936712-546). Unused canisters at the time of disposal shall not be a Resource Conservation Recovery Act (RCRA) hazardous waste as defined in 40 CFR 261.21 - 261.24 or 40 CFR 261.33 (e) and (f).

3.3 Absorbent. The absorbent shall conform to EA-C-1704.

3.4 Physical characteristics.

3.4.1. Canister size. The canister shall be able to fit inside, and be retained by, the canister carrier shell and cover (Drawings 5-1-1517 and 5-1-1515, respectively) used with the M42A1 Protective Mask. The maximum diameter of the canister shall be 4.20 inches, except for the top air inlet portion of the canister which extends above the carrier shell. This top portion shall have a diameter not greater than 4.40 inches. Canister height as measured from

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the canister inlet port (see 3.5.2 and figure 1) to the bottom air outlet port (see 3.5.2 and figure 2), shall be 3.00 ± 0.05 inches and represent the maximum overall height of the canister. The required overall canister height ensures proper interface with the motor blower assembly unit for the M43 series mask.

3.4.2 Weight. Canister weight shall not exceed 280.0 grams when tested as specified in 4.3.6.1.

3.4.3 Center of mass. The center of mass of the canister shall be not greater than 1.7 inches when measuring from the base of the outlet port along the canister axis (see 4.3.6.2).

3.4.4 Aerosol filter protective cover. The canister shall be equipped with a protective cover over the aerosol filter. The cover shall be capable of preventing punctures of the aerosol filter through the inlet port. The base of the cover shall have a diameter of not less than 1.26 inches and shall be located concentric to, and not less than 0.38 inches below, the inlet port as shown in figure 1. The protective cover shall not impair the connection of the armor quick disconnect assembly (Drawing 5−1−1960) to the canister inlet port or impede air flow when the armor quick disconnect assembly is used. The cover shall not crack or puncture when tested as specified in 4.3.6.3.

3.4.5 Color. The color of the canister shall be a uniform, flat or lusterless, dark, forest green (see 6.13).

3.5 Interface.

3.5.1 Canister air outlet port. The canister shall have a male threaded connector at the air outlet port that is concentric with the axis of the canister to allow attachment of the canister to the protective mask. The canister thread profile and dimensions of the connector shall conform to figure 2 (see 6.4).

3.5.2 Air inlet port design. The air inlet port shall be concentric to the axis of the canister. The air inlet port diameter, lip thickness and inside clearance shall conform to figure 1 (see 6.3).

3.6 Performance characteristics.

3.6.1 Absorbent moisture content. The average moisture content of the absorbent from completed canisters shall be not greater than 3.0 percent by weight and for any single canister shall be not greater than 5.0 percent by weight when tested as specified in 4.3.6.4.

3.6.2 Canister leakage. Canister leakage shall be not greater than 1 milliliter per minute when subjected to a vacuum of 75 millimeters (3 inches) of water created inside the canister when tested as specified in 4.3.6.5.

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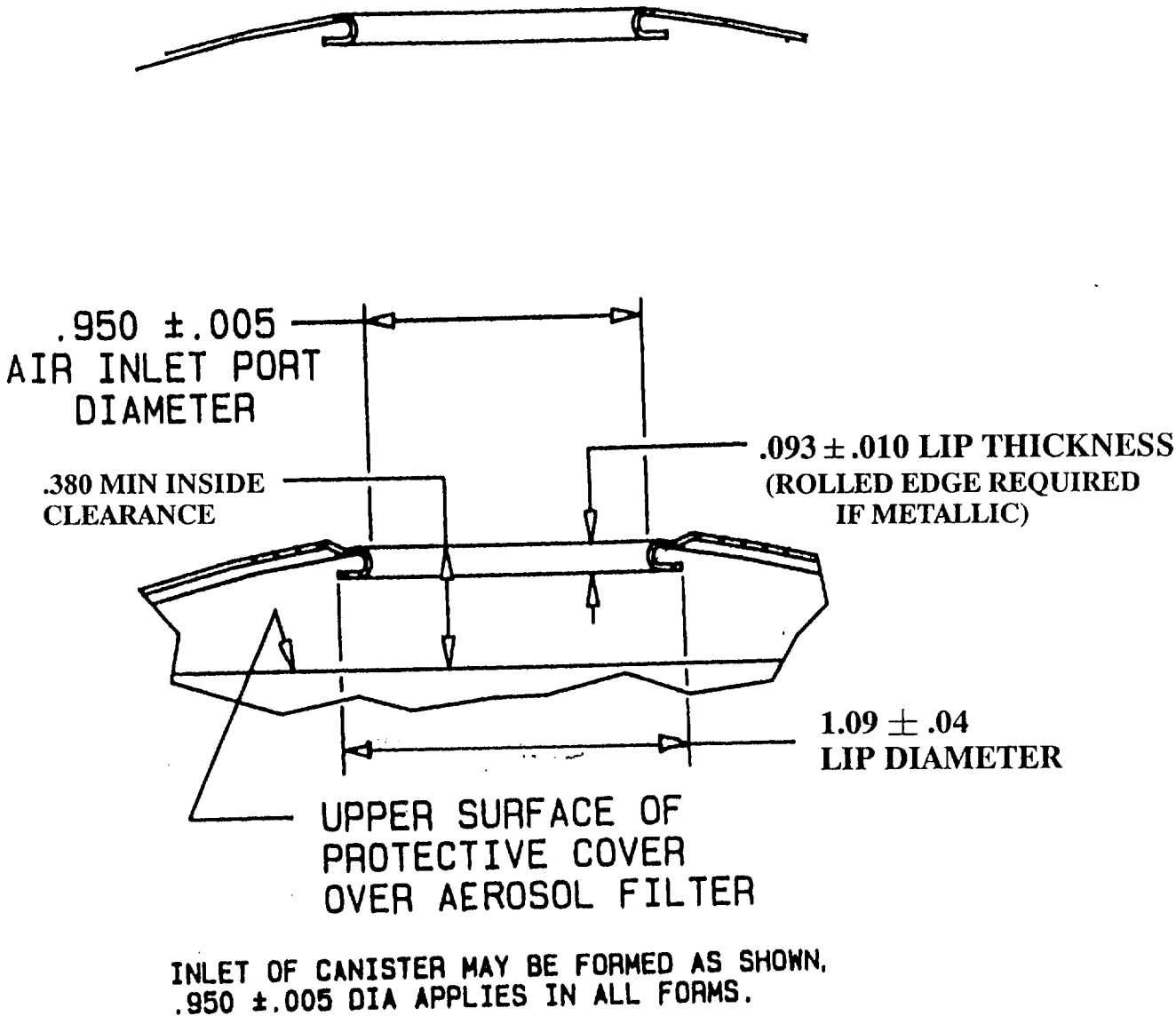


Figure 1. Air Inlet Port Dimensions

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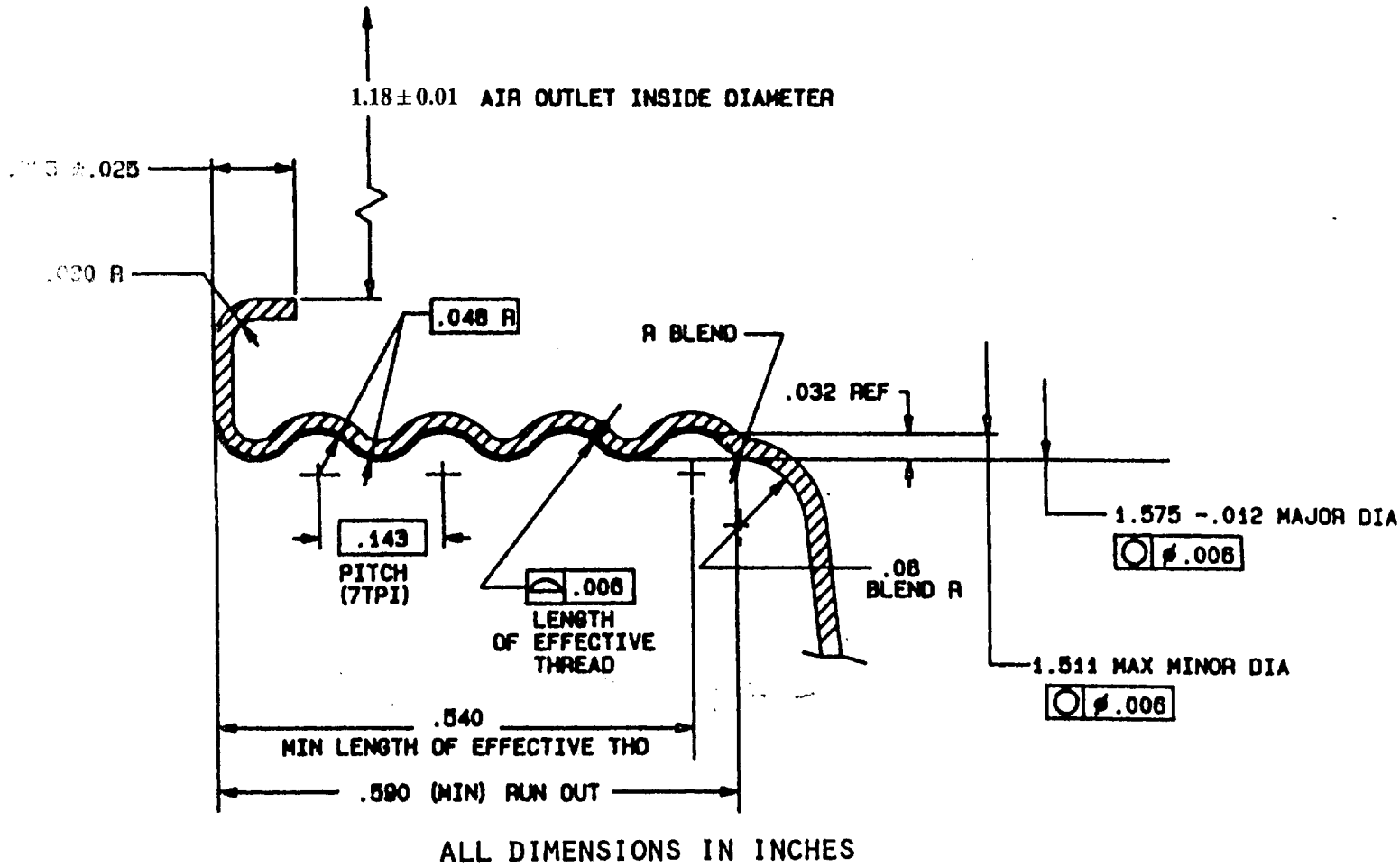


Figure 2. Canister Screw Thread (Bolt) Profile and Dimensions

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3.6.3 Airflow resistance. The airflow resistance of each assembled canister shall be not less than 10 millimeters of water nor greater than 17 millimeters of water at an airflow rate of 32 liters per minute The airflow resistance of each assembled first article canister shall be not greater than 45 millimeters of water at an airflow rate of 85 liters per minute when tested as specified in 4.3.6.6.

3.6.4 Aerosol filtration efficiency. The aerosol filtration efficiency of each assembled canister shall be not less than 99.990 percent (penetration of not greater than 0.010 percent) when tested as specified in 4.3.6.7 using particle smoke at a concentration of 100 ± 20 micrograms per liter, an average particle diameter of 0.3 micrometers and a flow rate of 32 liters per minute.

3.6.5 Particulate emission. The average particulate emission from the canister shall be not greater than 2 milligrams and shall be not greater than 5 milligrams from any individual canister when tested as specified in 4.3.6.8.

3.6.6 Liquid agent permeation. The canister shall be impermeable to mustard agent for not less than 360 minutes when tested as specified in 4.3.6.9.

3.6.7 Particulate filter clogging. After clogging the canister with dust, the airflow resistance of the canister shall increase by no more than 60 percent when tested as specified in 4.3.6.10.

3.6.8 Gas life. The calculated minimum gas life (the mean minus 1.1 times the standard deviation) of the canister shall be as specified in table I when tested as specified in 4.3.6.11.

TABLE I. Calculated gas life

Test agent	Minimum gas life, minutes (as manufactured)
Sarin (GB) 1/ 2/	83.0
Cyanogen chloride (CK) 3/	30.0
Dimethylmethyl–phosphonate (DMMP) 2/	59.0

1/ GB gas life shall be tested by the Government.

2/ GB and DMMP gas life – The break point shall be the time needed to attain an effluent concentration of 0.04 mg/m^3.

3/ CK gas life – The break point shall be the time needed to attain an effluent concentration equal to 8.0 mg/m^3.

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3.6.9 Decontamination capability. The exterior surface of the canister shall be capable of being decontaminated when exposed to liquid mustard (HD) agent as specified in 4.3.6.12.

3.7 Environmental conditions.

3.7.1 Rough handling. The canister shall meet the requirements of table II after being subjected to rough handling (see 4.3.6.13).

3.7.2 Accelerated aging. The canister shall meet the requirements of table II after being subjected to accelerated aging (see 4.3.6.14).

3.7.3 Environmental storage. The canister shall show no evidence of structural damage and shall meet the requirements of table II after being subjected to environmental storage (see 4.3.6.15).

3.7.4 Water repellency after accelerated aging. The average water repellency of each particulate filter assembly after accelerated aging (see 4.3.6.14) shall be not less than 300 millimeters of water when tested as specified in 4.3.6.16.

3.7.5 Canister integrity and performance after solar radiation/temperature cycling/rough handling. The canister shall have no cracks or material chips (removal of protective coating is permitted) and meet the requirements of table II after being subjected to solar radiation, temperature cycling and rough handling as specified in 4.3.6.17.

3.7.6 Corrosion resistance.

3.7.6.1 Caustic impact. After environmental storage (4.3.6.15) and subsequent exposure to carbonyl chloride (CG), canister leakage as a result of corrosion shall be not greater than one milliliter per minute when tested as specified in 4.3.6.18.1.

3.7.6.2 Salt spray. The canister shall show no sign of corrosion after being exposed to salt spray as specified in 4.3.6.18.2. If the canister is coated in order to meet the corrosion resistance requirement, no coating shall be removed from the canister when subjecting the canister to the wet tape adhesion test specified in 4.3.6.18.3.

3.7.7 Mildew resistance. There shall be no growth of the test microorganisms discernible to the unaided eye on any aerosol filter media of the canister when qualitatively tested as specified in 4.3.6.19.

3.7.8 Canister integrity after cold soak/bare drop. The canister shall not shatter or crack after exposure to the drop specified in 4.3.6.20.

3.8 Resistance to thermal radiation. The canister shall not melt, burn or sustain combustion when a thermal pulse of 15 calories per square centimeter is applied for 1 second as specified in 4.3.6.21.

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TABLE II. Performance characteristics after environmental conditioning

Characteristic	Minimum	Maximum	Test method
Airflow resistance, millimeters of water*			4.3.6.6
After rough handling	---	17	4.3.6.13
After accelerated aging	---	17	4.3.6.14
After environmental storage	---	17	4.3.6.15
Particulate filtration efficiency, percent**			4.3.6.7
After rough handling	---	0.025	4.3.6.13
After accelerated aging	---	0.010	4.3.6.14
After environmental storage	---	0.010	4.3.6.15
After solar radiation/temperature cycling/ rough handling	---	0.010	4.3.6.17
GB gas life, minutes			4.3.6.11
After rough handling	83.0	---	4.3.6.13
After accelerated aging (average)	83.0	---	4.3.6.14
After environmental storage (average)	75.0	---	4.3.6.15
DMMP gas life, minutes			4.3.6.11
After rough handling	59.0	---	4.3.6.13
After accelerated aging (average)	59.0	---	4.3.6.14
After environmental storage (average)	53.0	---	4.3.6.15
After solar radiation/temperature cycling/ rough handling	53.0	---	4.3.6.17
CK gas life, minutes			4.3.6.11
After environmental storage (average)	27.0	---	4.3.6.15
Particulate emission, milligrams			
After environmental storage			4.3.6.15
Average	---	2	4.3.6.8
From individual canister	---	5	4.3.6.8
Canister leakage, milliliter per minute			4.3.6.5
After environmental storage	---	1	4.3.6.15

*Airflow rate shall be at 32 liters per minute.

**Particle smoke or approved equal shall be used at at concentration of 100 ± 20 micrograms per liter, an average particle diameter of 0.3 microns and a flow rate of 32 liters per minute.

3.9 Resistance to battlefield contaminants. The canister shall show no evidence of cracks, material chips or degradation of its exterior after being exposed to battlefield contaminants and rough handling as specified in 4.3.6.22.

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3.10 Resistance to blast. The particle smoke penetration of the canister shall be not greater than 0.025 percent after exposure to a dynamic air pressure of 0.2 kilograms per square centimeter as specified in 4.3.6.23.

3.11 Marking. The area of the canister adjacent to the threaded connector shall be permanently marked, by means such as stamping or embossing, with the nomenclature "CANISTER, C2A1", National Stock Number and the lot number in characters approximately 0.125 inches high (see 6.2(e)). In lieu of stamping or embossing, the lot number may be marked using any opaque, nonfading, fast-drying, weather resistant stencil ink, lacquer, paint, or enamel. The markings shall be clear, legible, durable, nonfading and provide a definite contrast (such as white or yellow) to the canister surface.

3.12 Connector thread defects. The canister connector shall show no discontinuity or structural damage such as cracks, splits, tears, pitting, shearing or cuts when viewed with 20 power magnification as specified in 4.3.6.24.

3.13 Government-furnished property. The Government will furnish the following property to the contractor (see 6.5):

(a)	Decontamination Kit, Individual Equipment, M295	NSN 6850-01-357-8456
(b)	Paper, Chemical Agent Detector, M9	NSN 6850-01-226-5589
(c)	Canister Carrier Assembly, Combat Vehicle	NSN 4240-01-260-8704
(d)	Disconnect, Armor Quick Assembly	NSN 4730-01-392-2173

3.14 Workmanship. Canisters shall be free of sharp edges; foreign matter such as dirt, grease and oil; and damage such as cracks, dents, burrs, pitting, tears and scratches.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in 4.2.3 or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for

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compliance with any or all of the requirements of the applicable drawings, figures and special packaging instructions.

4.2.2 Sample. The first article sample shall consist of 600 canisters manufactured using the same methods, materials, equipment, and processes as will be used during regular production. Each canister shall be individually coded. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.3 Inspection procedure. Each sample canister shall be examined in accordance with the classification of characteristics (4.3.5). Sample first article canisters shall be tested as specified in table III. Sampling in accordance with table III shall be random. Each test result shall be recorded and identified with the canister number or code. Twenty "as manufactured" canisters, unit packed as specified in the contract, shall be forwarded to the Government laboratory for GB gas life testing in accordance with 4.3.6.11.

4.2.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the canisters produced by one manufacturer, at one plant, from the same materials, under the same manufacturing conditions, except that no more than one lot of absorbent, one lot of particulate filter.media or one week's production shall be represented in any one lot of canisters. Any change in equipment, materials, or procedure used in regular production from that used in first article shall be approved by the Government prior to its institution. The lot numbers shall be assigned according to normal industry practice (see 6.7).

4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with the classification of characteristics in 4.3.5 and, when specified, table IV. Samples shall be selected at random.

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TABLE III. First article tests

Test type	Number of items						
	As manu–factured	After rough handling	After accelterated aging	After environ–mental storage(D)	After solar radiation	After bare drop	Test method
Airflow resistance	600	20(B)	40(C)	165(E)	---	---	4.3.6.6.1
Particulate filtration efficiency	600	20(B)	40(C)	165(E)	20(H)	---	4.3.6.7
Absorbent moisture content	15(A)	---	---	---	---	---	4.3.6.4
Canister connector defects	15(A)	---	---	---	---	---	4.3.6.24
Mildew resistance	8(A)	---	---	---	---	---	4.3.6.19
Particulate emission	50	---	---	45(E)(F)	---	---	4.3.6.8
Water repellency (filter)	---	---	20(C)	---	---	---	4.3.6.16
Particulate filter clogging	15	---	---	---	---	---	4.3.6.10
CK gas life	20	---	---	60(E)	---	---	4.3.6.11
DMMP gas life	20	20(B)	20(C)	60(E)	20(H)	---	4.3.6.11
GB gas life	20(G)				---	---	4.3.6.11
Visual inspection for damage (package, unit pack and canister)	---	---	---	165	---	---	---
Canister leakage	600	---	---	45(E)(F)	---	---	4.3.6.5
Corrosion resistance (caustic impact)	---	---	---	45(E)(F)	---	---	4.3.6.18.1

(A) Same canisters may be used for charcoal loss in weight, canister connector defects and mildew resistance.

(B) Same canisters may be used for after rough handling tests of airflow resistance, particle smoke penetration and DMMP gas life.

(C) Following the after aging tests for airflow resistance and particle smoke penetration, the same canisters may be used for water repellency or DMMP gas life after aging.

(D) Sample shall contain an equal number of canisters from each of three environmental storage conditions (see 4.4.6.15).

(E) Same canisters may be used for the after environmental storage tests of airflow resistance and particle smoke penetration. The canisters used for after environmental storage tests of CK gas life, DMMP gas life, canister leakage, charcoal dust emission and corrosion resistance may come from the sample of canisters for airflow resistance and particle smoke penetration after environmental storage.

(F) Same canisters may be used for after environmental storage tests of charcoal dust emission, canister leakage and corrosion resistance. After environmental storage, canisters shall be tested for charcoal dust emission prior to airflow resistance, particle smoke penetration and canister leakage.

(G) Test to be conducted by the Government (see 6.6)

(H) Same canister may be used for after solar radiation tests of particulate filtration or DMMP gas life.

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TABLE III. First article tests – continued

Test type	Number of items						Test method
	As manu-factured	After rough handling	After accelerated aging	After environ-mental storage	After solar radiation	After bare drop	
Weight	600	---	---	---	---	---	4.3.6.1
Center of mass	20	---	---	---	---	---	4.3.6.2
Inlet port protective cover	8	---	---	---	---	---	4.3.6.3
Liquid agent permeation	20(I)	---	---	---	---	---	4.3.6.9
Decontamination capability	20	---	---	---	---	---	4.3.6.12
Corrosion resistance salt spray	15	---	---	---	---	---	4.3.6.18.2
Adhesion(J)	15	---	---	---	---	---	4.3.6.18.3
Canister integrity and performance after solar radiation/temp. cycling	---	20	---	---	---	---	4.3.6.17
Resistance to thermal radiation(K)	3	---	---	---	---	---	4.3.6.21
Resistance to battlefield contaminants(L)	---	38	---	---	---	---	4.3.6.22
Resistance to blast(K)	3	---	---	---	---	---	4.3.6.23
Canister integrity after cold soak	---	---	---	---	---	15	4.3.6.20

(I) Eight 6-by-6 inch specimens shall be fabricated from external canister material

(J) Test required if canister is coated.

(K) Tests to be conducted by the Government (see 6.6).

(L) Two canisters per contaminant.

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TABLE IV. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.3.2.2 For test. Sampling shall be conducted in accordance with table V. Samples shall be selected at random.

4.3.3 Inspection procedure.

4.3.3.1 For examination. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items shall be examined in accordance with the classification of characteristics in 4.3.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.3.3.2 For test. Sample canisters shall be tested as specified in table V. If any sample item fails any of the tests specified in table V, the lot represented shall be rejected, unless otherwise noted in the table and corresponding footnotes.

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TABLE V. Conformance tests

Test type	Number of samples		
	As manufactured	After rough handling	Test method
Airflow resistance	100 percent; Acc 0; Rej 1 (A)	20 (B)	4.3.6.6.2
Particulate filtration efficiency	100 percent; Acc 0; Rej 1 (A)	20 (B)	4.3.6.7
Weight	Table IV, Level VII	----	4.3.6.1
Canister leakage	Table IV, Level VII	----	4.3.6.5
Absorbent moisture content	8 (C)	----	4.3.6.4
Canister thread defects	8 (C)	----	4.3.6.24
Charcoal dust emission	50	----	4.3.6.8
CK gas life	20 (D)	----	4.3.6.11
DMMP gas life	20 (D)	20 (B)(D)	4.3.6.11

(A) – Any canister that fails particulate efficiency or airflow resistance "as manufactured" shall be rejected.
(B) – Same canisters may be used for airflow resistance, particle smoke penetration and DMMP gas life tests after rough handling.
(C) – Same canisters may be used for absorbent moisture content and canister thread defects tests.
(D) – If the calculated gas life for the sample canisters fails to meet the requirements of table I, the lot of canisters shall be rejected.

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

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4.3.5 Classification of characteristics. Conformance examinations shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table IV for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.5(a)	Unpacked canisters		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Overall dimensions correct	Table IV, Level VII	3.4.1	CE
102	Canister height correct	Table IV, Level VII	3.4.1	CE
103	Air inlet port diameter correct	Table IV, Level VII	3.5.2	CE
104	Air inlet port lip thickness correct	Table IV, Level VII	3.5.2	CE
105	Air inlet port inside clearance correct	Table IV, Level VII	3.5.2	CE
106	Rolled thread major diameter correct	Table IV, Level VII	3.5.1	CE*
107	Rolled thread minor diameter correct	Table IV, Level VII	3.5.1	CE*
108	Effective thread length correct	Table IV, Level VII	3.5.1	CE*
109	Air outlet inside diameter correct	Table IV, Level VII	3.5.1	CE
110	Workmanship	Table IV, Level VII	3.14	VI
111	Marking correct, evident, and legible	Table IV, Level VII	3.11	VI
Minor				
201	Color	Table IV, Level IX	3.4.5	VI

NOTES:

*See 6.8
CE – Commercial inspection equipment
VI – Visual inspection

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4.3.6 Tests.

4.3.6.1 Canister weight. Canister weight shall be determined using a balance accurate to 0.1 gram.

4.3.6.2 Center of mass. Using a surface plate, place the canister in a "V" block. Locate center of canister, with height gage, and scribe a light line on both sides. Set canister on threaded end and, by starting near top of the canister and moving toward threaded end, scribe a series of perpendicular graduated lines 0.062 inch apart. Locate canister between a set of bench centers by using the center line and graduated lines. Mark point at which the center of mass is reached. Place the canister on surface plate and measure the distance the center of mass point is from the threaded end of the canister.

4.3.6.3 Aerosol filter protective cover. The following apparatus shall be used: (1) Base: Plywood approximately 14−by−14−by−3/4 inches (length, width, thickness) (2) Vertical column: wooden column secured perpendicular to the base. The column supporting a 1.250 inch diameter aluminum tube having an inside diameter of 1.075 inches and a length of 46 inches. The tube located 6 inches from the base. (3) Holding block: Hardwood block 6−by−6−by 1−inch (length, width, thickness) with a 1 inch diameter hole through to receive the threaded end of the canister. (4) Projectile: 1−inch diameter tool steel projectile weighting 0.10 pounds with a 0.125 inch diameter radiused 60° included angle. Using this apparatus, the tool steel projectile shall be dropped through the tube onto the protective aerosol cover of the canister which is centered below the tube.

4.3.6.4 Absorbent moisture content. The absorbent moisture content shall be determined in accordance with ASTM D 2867, Oven−Drying test method, except the oven temperature shall be 103° to 107°C and the drying time shall be 3 hours.

4.3.6.5 Canister leakage. Canister leakage shall be determined with the Q204 Air Leakage (Dry Bubble) Tester (Drawing 136−301−38) or approved equal.

4.3.6.6 Airflow resistance.

4.3.6.6.1 First article inspection. The airflow resistance at 32 liters per minute shall be determined with the Q127 Filter Testing Penetrometer (Drawing 136−42−102) or approved equal using a chuck gasket with a durometer hardness of A/50 to A/55. The airflow resistance for manufactured canisters at 85 liters per minute shall be determined with the Q112 canister resistance indicator (Drawing 38−6−351−8) or approved equal.

4.3.6.6.2 Conformance inspection. The airflow resistance at 32 liters per minute shall be determined with the Q127 Filter Testing Penetrometer (Drawing 136−42−102) or approved equal using a chuck gasket with a durometer hardness of A/50 to A/55.

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4.3.6.7 Aerosol filtration efficiency. The aerosol filtration efficiency at 32 liters per minute shall be determined with the Q127 Filter Testing Penetrometer (Drawing 136−42−102) or approved equal using a chuck gasket with a durometer hardness of A/50 to A/55.

4.3.6.8 Particulate emission. The dust emission shall be measured using the Q261 Canister Shaker Tester (Drawing 136−45−842) or approved equal.

4.3.6.9 Liquid agent permeation. The liquid agent permeation test shall be conducted in accordance with appendix A.

4.3.6.10 Particulate filter clogging. After the canister has been tested for the "as manufactured" airflow resistance, the canister shall be exposed to blowing dust of 200 mesh silica dioxide flour in a test chamber maintained at a temperature of 26 ± 3°C and a relative humidity of not greater than 30 percent. Place the canister in the chamber with the inlet port facing directly into the blowing dust. Pull air through the canister while subjecting the canister to blowing dust with a wind velocity of 1750 feet/minute and a dust density maintained at 40 percent opacity (the device used to measure dust density shall be calibrated to a number 2 Ringleman test screen). The net airflow through the canister shall be 80 ± 10 liters per minute. After the canister has been exposed to the specified blowing dust for not less than 10 minutes, the airflow resistance shall be determined in accordance with 4.3.6.6.2 and the percent increase calculated. NOTE: Silica dioxide fluor is a suspected carcinogen. Respiratory protection shall be worn when handling this material.

4.3.6.11 Gas life. The CK gas life shall be determined with the Q95 All Purpose Gas Life Testing Apparatus (Drawing 136−41−600−1) or approved equal. The DMMP and GB gas life of the canister shall be determined with the Q250 Fifty Liter Per Minute Filter Life Tester (Drawing 136−41−3500) or an approved equal gas life testing apparatus. Gas life conditions shall be as shown in table VI.

4.3.6.12 Decontamination capability. The exterior of the canister shall be exposed to unthickened distilled mustard (HD) drops having a mass median diameter of 1.4 ± 0.16 millimeters until a contamination density of 10 ± 1 gram/square meter is achieved. The mustard shall remain on the surface for not less than 1 hour. After the 1 hour period, the canister shall be decontaminated with the M295 Individual Equipment Decontamination Kit using standard field procedures. After completing the decontamination operation, the M9 Chemical Agent Detector Paper shall be used to determine if any residual mustard is detectable. An M9 paper color change ranging from pink−red to red shall indicate the presence of mustard and represent failure of this test.

4.3.6.13 Rough handling. Canisters shall be rough handled 10 minutes ± 5 seconds using the Q113 Rough Handling Apparatus (Drawing 26−19−51), or approved equal, equipped with the roll bar. No more than four canisters at a time shall be rough handled. The canister thread shall

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TABLE VI. Conditions for gas life testing

Gas	Concentration	Temperature	Relative humidity(C)	Rate of airflow
GB, (90 percent min purity) (A)	4000 ± 400 mg/m ^{3}	26 ± 3° C	Not more than 15%	32 liters/ min (D)
CK, (technical grade) (B)	4000 ± 200 mg/m ^{3}	26 ± 3° C	80 ± 3%	32 liters/ min (E)
DMMP, (96 percent min purity) (A)	3000 ± 200 mg/m ^{3}	26 ± 3° C	Not more than 15%	50 liters/ min (D)
CG, (98 percent min purity)	20000 ± 500 mg/m ^{3}	26 ± 3° C	50 ± 3%	32 liters/ min (D)

(A) GB and DMMP gas life − The break point shall be the time needed to attain an effluent concentration equal to 0.04 mg/m^3 (see 6.9)
(B) CK gas life − The break point shall be the time needed to attain an effluent concentration equal to 8.0 mg/m^3.
(C) Canisters for CK gas life testing and CG corrosion resistance shall be equilibrated at 80 ± 3% relative humidity. Other canisters are gas tested as−received.
(D) Constant flow
(E) Intermittent flow

be protected by a protective cap. Immediately following rough handling, the canister shall be sealed with a cap and plug to protect against moisture adsorption (see 6.10).

4.3.6.14 Accelerated aging. Canisters, unit packed as specified in the contract, shall be aged in a forced−air circulating oven for 120 hours at 117 ± 3°C.

4.3.6.15 Environmental storage. A sample of 55 canisters, unit packed in accordance with the contract, shall be stored at each of three environmental conditions for 12 weeks. Storage conditions shall be hot dry (+71°C), cold (−51°C) and cyclic (one week hot−dry, one week cold, and one week hot−wet at 45°C, 85 percent relative humidity) repeated for four cycles. After environmental storage, the unit packs and canisters shall be tested in accordance with table II.

4.3.6.16 Water repellency of the particulate filter assembly. After canister accelerated aging (4.3.6.14), particulate filters shall be removed from canisters for water repellency testing. No canister exposed to gas tests shall be used. Care shall be taken to prevent damage to filters during the removal process. Filters showing any indication of damage due to the removal process shall be discarded. The water repellency of the filter assemblies shall be determined using the Q101 Water Repellency Indicator (Drawing 125−8−1−2) or approved equal. No indicator paper can be used for this test. In the absence of the indicator paper, the operator shall visually determine when the water has penetrated the filter assembly.

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4.3.6.17 Canister integrity and performance after solar radiation/temperature cycling/ rough handling. The canister shall first be subjected to solar radiation in accordance with Appendix B (see 6.12). After exposure to solar radiation, the canisters shall be prepared for temperature cycling as follows: Place a skin thermocouple on one of the canisters. Store the test canisters in an environmental chamber set to −25°F. After the skin thermocouple reaches −25 ±2°F, soak the canisters for a minimum of 4 hours. Remove the canisters and, within 5 minutes, place the test canisters in an environmental chamber at 160°F. After the skin thermocouple reaches 160 ±2°F, soak the canisters for a minimum of 4 hours. Remove the canisters and, within 5 minutes, return the canisters to the −25°F chamber. Repeat this procedure until the canisters have been in the cold chamber four times and in the hot chamber three times. After the last cold chamber exposure, allow the canisters to return to ambient temperature and inspect. Finally, rough handle the canisters as specified in 4.3.6.13, visually examine for damage and test for compliance to the requirements of table II (solar radiation exposure).

4.3.6.18 Corrosion resistance.

4.3.6.18.1 Caustic impact. After environmental storage (4.3.6.15), canisters shall be exposed to CG using the Q95 All Purpose Gas Life Testing Apparatus or approved equal. Exposure shall be for 5 ± 0.1 minutes using conditions shown in table VI. After exposure, canisters shall be immediately sealed using a cap and plug (see 6.10). After storage for 28 days at ambient conditions (26 ± 3°C) canisters shall be leak tested as specified in 4.3.6.5 (see 6.11).

4.3.6.18.2 Salt spray. The canister shall be subjected to salt spray in accordance with ASTM B 117 for 96 hours (48 hour exposure followed by a 48 hour drying period). Upon completion of the drying period the canister shall be cut in half axially and examined for corrosion both on the exterior and interior of the canister.

4.3.6.18.3 Adhesion. The canister (with plugged port and capped threads) shall be immersed in distilled water for 24 hours. The canister shall be removed and dried. Within 1 minute of removal from the water, two parallel scratches, one inch apart, shall be made through the coating with a razor blade, scalpel, knife or other cutting device. A one−inch wide strip of masking tape, with the adhesive side down, shall immediately be applied across the scratches. The tape shall have an average adhesion of 60 ounces per inch of width. Using a roller, the tape shall be firmly pressed against the surface of the coating while leaving no air bubbles between the coating and the tape. Within 90 seconds of application, the tape shall be removed with one quick motion (not jerked), pulling the tape back upon itself at as close to an 180° angle as possible. The area shall be examined for paint removal.

4.3.6.19 Mildew resistance. Mildew resistance shall be determined in accordance with method A of ASTM D 2020.

4.3.6.20 Canister integrity after cold soak/bare drop. The canister shall be cold soaked at −50°F for not less than 1 hour. Immediately thereafter the canister shall be dropped from a height of no less than 4 feet onto a 2−inch thick plywood surface. The canister shall be dropped

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three times with three different impact orientations. The canister shall be dropped with the threads face down, side facing down and the inlet port facing down.

4.3.6.21 Resistance to thermal radiation. The canister shall be subjected to thermal radiation using test equipment and the test operating procedures of the U.S. Army White Sands Missile Range, New Mexico (see 6.6).

4.3.6.22 Resistance to battlefield contaminants. Each test sample shall be sanitized by washing with mild detergent, rinsing and air drying. The test apparatus shall be in accordance with figures 3 and 4. Materials used to construct the apparatus shall be compatible with the contaminants being used. Separate samples shall be contaminated with each of the compounds specified in table VII. The sample shall be marked to indicate the surface to be contaminated and the direction of contaminant flow. The canister shall be mounted on the test apparatus of figure 3 at an angle of 45° to the spray head with inlet port facing downward for contamination and draining. The spray nozzle shall be adjusted horizontally so that the spray pattern hits near the center of the top edge of the sample without splashing over the edge. The spray head shall be adjusted vertically so that the distance between the center of the nozzle face and the surface of the test specimen is 6 inches. Samples shall be contaminated uniformly using the spray nozzle through which a 50 milliliter volume of contaminant has been funnelled at room temperature. The contaminant shall be poured into the funnel quickly, but steadily, so that spraying shall be continuous once it has begun. The sample shall be allowed to drain for 15 minutes, the surface carefully blotted dry with a paper towel, and then air dried for an additional 30 minutes. The sample shall be washed with a mild detergent, rinsed and air dried for 30 minutes. The sample canister shall then be subjected to rough handling as specified in 4.3.6.13.

4.3.6.23 Resistance to blast. The canister shall be subjected to dynamic air pressure using test equipment and the test operating procedures of the U.S. Army White Sands Missile Range, New Mexico (see 6.6).

4.3.6.24 Connector thread defects. Using 20 power magnification, the sample canister shall be completely examined around the entire circumference of the threaded area for shearing, damage, cracks, cuts, splits, pitting and tears. Using a jeweler's saw or high speed cutting device to avoid damaging the thread configuration, the sample canister shall then be carefully sectionalized perpendicular to the major diameter of the canister thread. The first cut shall be made at the point where the thread begins and three other cuts shall be made 90° apart. The cut areas shall be ground to remove the saw cut metal. Under magnification, the threaded area shall be examined for the specified defects.

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Table VII . Battlefield contaminants

Contaminant	Concentration
gasoline	Neat
diesel fuel	Neat
F-40 turbine fuel	Neat
Decontaminating Solution No. 2 (DS2)	*
Fichlor Decontaminating Solution (5 percent)	**
ethylene glycol	Neat
propylene glycol	Neat
PEG 200	Neat
methanol	Neat
ethanol	50/50 with water
Super Tropical Bleach (68 percent slurry)	40 parts STB to 60 parts water by weight
perchloroethylene	Neat
methyl ethyl ketone	Neat
trichloroethylene	Neat
insect repellents (NSN 6820-00-753-4963)	Neat
hydraulic fluid NATO O-148	Neat
hydraulic fluid NATO A-515	Neat
White Spirits	Neat
LSA (NSN 9150-00-935-6597)	Neat

*DS2 formulation (percent by weight): 70 percent diethylene triamine, 28 percent methyl cellusolve and 2 percent sodium hydroxide.

**Fichlor formulation (percent by weight): 5.0 percent sodium N,N-dichloroisocyanurate, 2.5 percent sodium hydroxide, 0.5 percent boric anhydride, 0.01 percent Santomerse detergent (alkyl benzene sulfonate), and remainder water.

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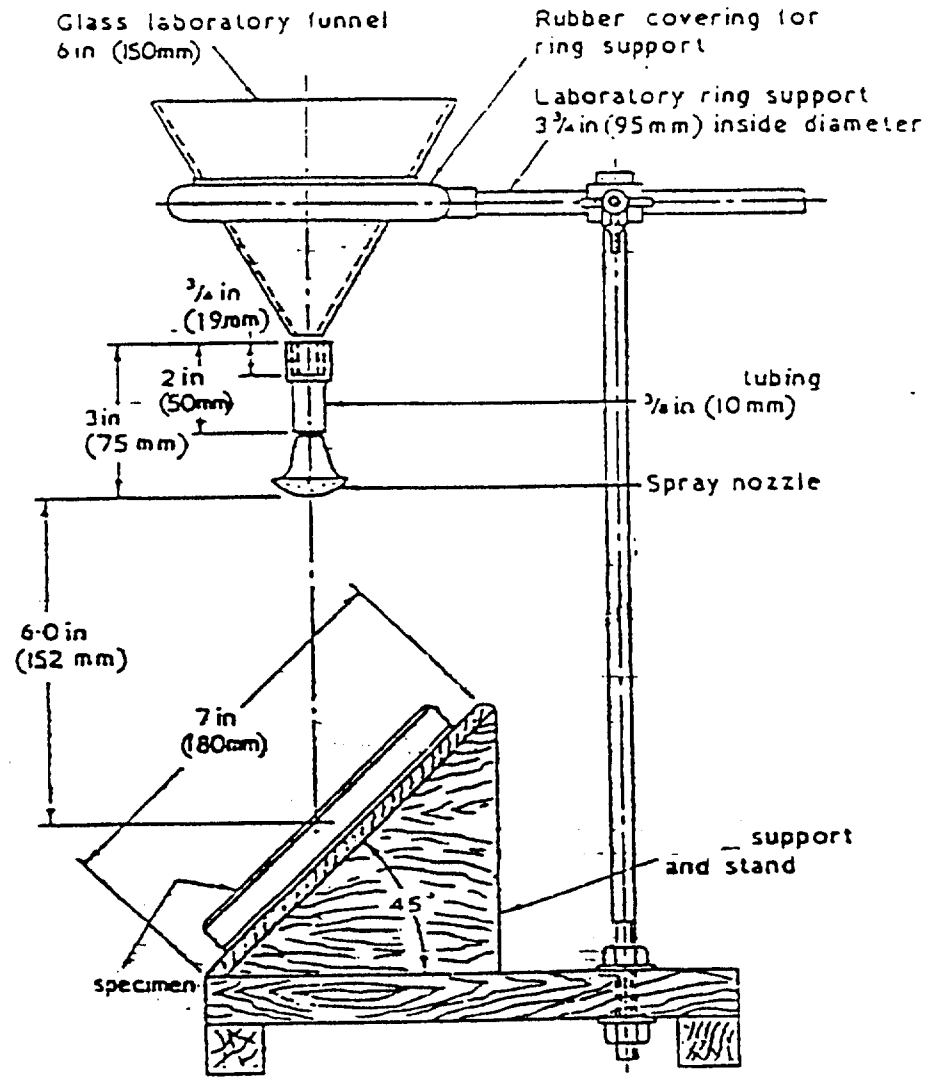


Figure 3. Apparatus for spraying battlefield contaminants

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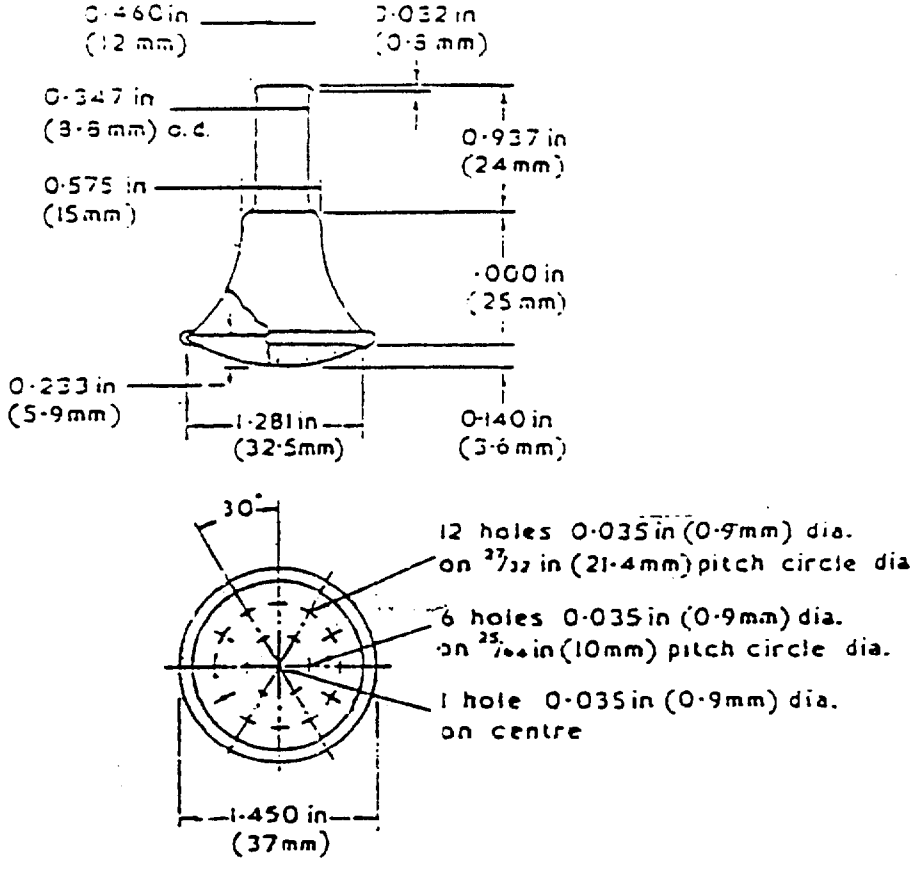


Figure 4. Spray nozzle

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers a canister which is intended for use as a component of the MCU−2A/P, M40, M42, M43, M45, M48, and M49 Protective Mask Systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see section 2).
(c) Packaging requirements (see 5.1).
(d) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production

(e) National Stock Number (see 3.11)

6.3 Air inlet port design. Control of the air inlet port dimensions is required to assure compatibility with the armor quick disconnect for the M42A1 Chemical−Biological Protective Mask.

6.4 International standardization agreements. Certain provisions of this specification (3.5.1) are the subject of international standardization agreements QSTAG 496 and QSTAG 695. When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate

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action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.5 Government−furnished property. The contracting officer should arrange to furnish the property listed in 3.13.

6.6 Samples for Government testing. Samples for Government testing for GB gas life should be forwarded to Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD−ENM−S, Building E5100, Aberdeen Proving Ground, MD 21010−5423. Samples for thermal radiation and nuclear blast testing should be forwarded to U.S. Army White Sands Missile Range, ATTN: STEWS−DATTS−AC, White Sands Missile Range, NM 88002.

6.7 Lot numbering. MIL−STD−1168 outlines a lot numbering procedure that has been used successfully during previous canister productions.

6.8 Canister thread dimensions. Rolled thread major diameter should be measured using a Go Rolled Thread Ring Gage (Drawing 5220−A51−2973) and a Not Go Plain Ring Gage (Drawing 5220−A51−2976). Effective thread length and rolled thread minor diameter should be measured using the Go Rolled Thread Ring Gage (Drawing 5220−A51−2973) only.

6.9 DMMP test reagent. DMMP may be purchased from Albright and Wilson, Americas, P.O. Box 26229, Richmond, VA 23260 or Aldrich Chemical Co., Inc., P.O. Box 2060, Milwaukee, WI 53201.

6.10 Caps and plugs. Caps and plugs used in conjunction with the corrosion resistance test (4.3.6.18) and the rough handling test (4.3.6.13) are required for test purposes only. Other caps and plugs capable of providing protection equivalent to those specified for use with the Q113 Rough Handling Apparatus (Drawing 26−19−51) may be used if approved by the Government.

6.11 Corrosion resistance. Phosgene adsorbed on humidified charcoal will be rapidly converted to hydrochloric acid. The corrosion resistance test is designed to measure the protection afforded by the coating on the interior of the canister shell against hydrochloric acid. If more convenient to the vendor, an equivalent molar weight of chloride in the form of hydrochloric acid can be substituted for the phosgene specified in 4.3.6.18. Care should be taken in conducting the corrosion test to assure no hydrochloric acid desorbs from the canister into the Q204 Air Leakage Tester. A protective filter between the test canister and the Q204 Air Leakage Tester may be necessary.

6.12 Solar radiation. Background information and a more thorough description of the test facilities, equipment and procedures used for solar radiation testing may be found in MIL−STD−810, Method 505.3.

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6.13 Color. The color of the canister should approximate color chip 34079 of FED-STD-595.

6.14 Technical data package. For information only, previous procurements of the C2A1 canister have used the following technical data package:

Drawing 5-3-1520 - Canister, Chemical-Biological Mask, (ASZM-TEDA) C2A1
MIL-C-51560, Amendment 4 - Canisters, Chemical-Biological Mask: C2 and C2A1

(Copies are available from Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-M, Aberdeen Proving Ground, MD 21010-5423.)

6.15 Subject term (key word) listing.

Carbon, activated
CK
Cyanogen chloride
Dimethylmethylphosphonate
Dioctylphthalate
DMMP
DOP
GB
Sarin

6.16 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A226

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APPENDIX A

MUSTARD RESISTANCE OF IMPERMEABLE MATERIALS (STATIC-DIFFUSION METHOD)

A.1 SCOPE

A.1.1 Scope. This appendix describes a static diffusion method for determining the mustard resistance of impermeable materials. Mustard resistance, when measured by this method, is the time for sufficient mustard to pass through the material, vaporize, and cause a "vapor break" as indicated on the indicator paper. It is not the time for the liquid film to pass through the material. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

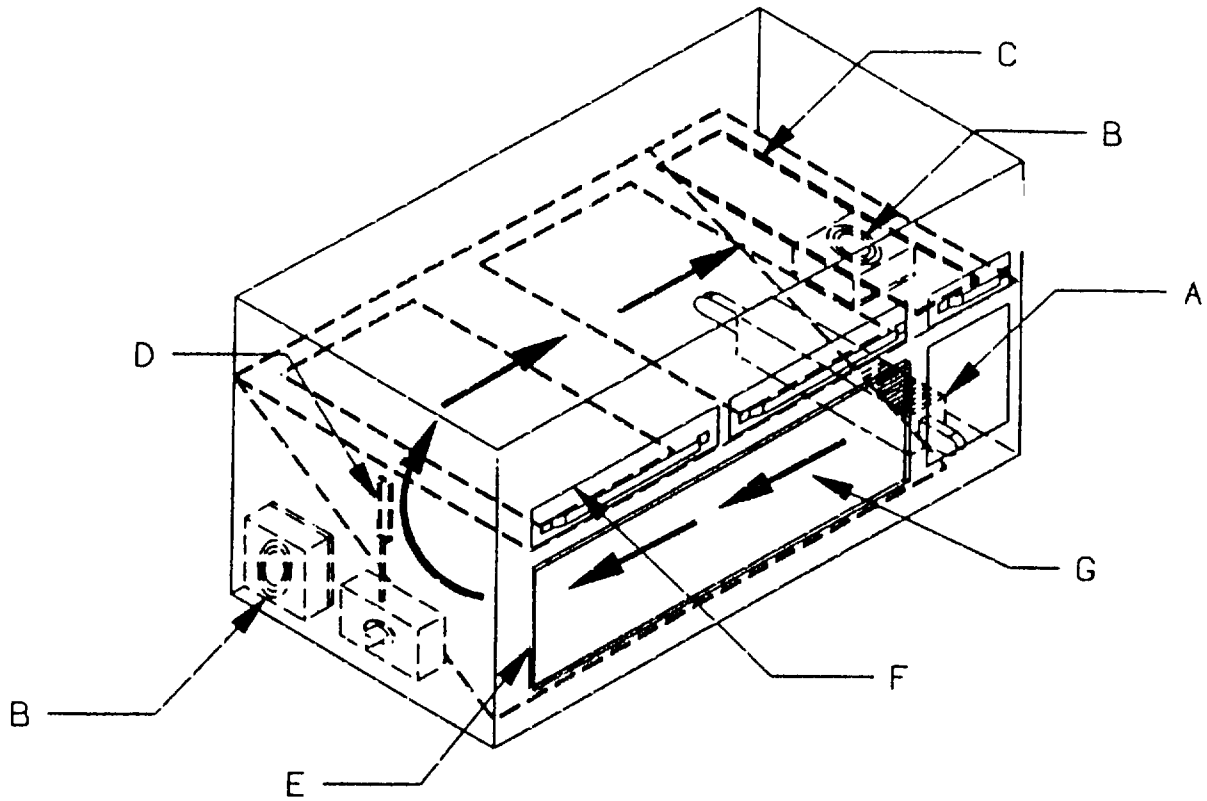
A.3 OUTLINE OF METHOD

A.3.1 The test is conducted by placing a specified quantity of liquid mustard (HD) on the top surface of the material, which rests on standard congo red indicator paper. When mustard permeates the test material, it evaporates and the vapors react with S-328 (tetrachloro-7, 8-diphenylglycoluril) that has been spotted on the paper, producing hydrogen chloride (HCl) that reacts with the impregnant congo red, changing it to a blue color. The time required for sufficient mustard to pass through the test material and effect the color change in the indicator paper is the mustard resistance of the material.

A.4 APPARATUS

A.4.1 The apparatus, Q170 (see figure 5), used in an oven thermostatted at 37° ± 0.5°C, containing a heater, circulating fan or fans, charcoal filter bed, drawers to hold glass trays of test specimens, and mirrors to view the undersides of the specimen trays. The specimen trays are made of glass plates that permit seeing the detector paper under the mounted specimens when viewed through the mirror, which is mounted at an appropriate angle underneath the trays. A glass window in the front wall of the oven allows the mirror to be seen. The drawers can be pulled part way out or completely removed to allow the trays of specimens to be placed on the drawers. The bottoms of the drawers are open so that the specimens can be seen through the glass trays. If the drawers are to be removed, dummy fronts can be inserted into the openings to maintain the internal temperature. The dimensions of the oven and the number of drawers can be tailored to fit the available hood space. The heater and thermostat are designed to maintain the internal temperature at 37° ± 0.5°C, with a relatively fast rise without overshooting the required temperature. The fan, or fans, circulate the air under the trays, through an opening at the end of the trays, across the top of the trays, and through a charcoal filter to the underside.

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A. Heating element
B. Fans
C. Bed of Charcoal
D. Themoregulator
E. Mirror
F. Drawers (each holds 20 samples)
G. Window

Figure 5. Static-diffusion apparatus

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The air circulation is required to maintain a uniform temperature, but the velocity is low enough not to disturb the specimens on the trays. The charcoal filter serves to adsorb any mustard vapor that might escape from the specimens. The filter is made of loose charcoal spread in a tray whose bottom is made of wire screen; a cloth scrim may be laid in the tray to prevent fines from passing through. The walls of the apparatus should be insulated to prevent heat loss.

A.5 PREPARATION FOR TEST

A.5.1 Indicator paper. (See A.8.3)

A.5.1.1 S-328 solution. Mix 0.5 gm of S-328 (tetrachloro-7, 8-diphenylglycoluril) and approximately 0.1 gm of anhydrous sodium or potassium carbonate with 10 mL of 1,1,2,2-tetrachloroethane. Shake the mixture thoroughly and filter through a No. 42 filter paper. Pour the clear filtrate into a small actinic glass bottle. The bottle should be kept closed at all times when not in use. A fresh solution should be prepared weekly.

A.5.1.2 Impregnating. Dip 4.25-cm circles of qualitative analytical filter paper for coarse, gelatinous or medium precipitates into approximately 0.1 percent solution of congo red in distilled water. When the paper is thoroughly impregnated, allow to air dry. The treated paper may be kept indefinitely but must be prepared and stored in an acid-free and mustard-free atmosphere.

A.5.1.3 Spotting. Spot the impregnated paper with the S-328 solution as follows: Dip a cotton swab or similar tool into the S-328 solution and then touch the swab to the congo red impregnated paper. Make one spot at the center of the paper and four others about it at a distance of approximately 1/4-inch. Lets the spots dry. The spotted paper is defective after 24 hours and should be used preferably before 12 hours have elapsed. If the test specimen runs more than 24 hours, the indicator paper should be changed at the end of the 24-hour period. Examine the dried spots before use; if a discoloration, even a faint one, appears, it is likely the S-328 solution is defective and should be remade before continuing with the test.

A.5.1.4 Apparatus. Turn on the fan and temperature controller. The temperature must be maintained at 37° ± 0.5°C, measured on the glass plate on which the specimens are placed. To simulate total mass that must be maintained at this temperature during the test, lay two glass plates in each tray, one to represent the cover glasses, and adjust the controller settings until the temperature can be maintained. These settings should be adequate for the duration of the test. If the apparatus is used on a daily basis, it is best to leave the power on at all times to maintain the temperature. If a temperature other than 37° ± 0.5°C is indicated bring to the correct temperature by adjusting the controller. If a variable temperature is found in the apparatus check the temperature regulating system for defects.

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A.5.1.5 Test specimens.

A.5.1.6 Cutting specimens. Cut the required number of disks or swatches from the material to be tested. The specimens should be slightly larger than the rings used to mount them (see A.5.1.7).

A.5.1.7 Mounting specimens. Place the specimens on a smooth, flat surface. Attach a ring that is impervious to mustard to the specimen and press firmly until it is securely set, providing an airtight seal between specimen and ring. A suitable method is as follows: Melt equal parts beeswax and crystalline paraffin wax. Immerse felt rings into the melt until the rings are saturated with wax. Remove a ring from the melt with tongs, allow excess liquid wax to drain, and place the ring on the specimen. Place a weight on the ring to hold it in place until the wax cools and the ring is firmly attached to the specimen. Do not allow any wax to coat the specimen inside the ring. The ring should have a 3/4 to 1-inch inside diameter and be approximately 1/4-inch high.

A.6 PROCEDURE

A.6.1 Preconditioning of specimens. Place disks of the spotted indicator paper (spotted side down) on a glass plate. Place the mounted specimens on the indicator-paper disks, with the specimen contacting the paper. Place the glass plate, with specimens and indicator paper into a clean oven at 37°C for enough time to allow them to attain 37°C. Also place glass cover plates into the oven for preconditioning at 37°C.

A.6.2 Testing. Set the preconditioned glass plate and specimens on a platform heated to 37°C in front of the testing apparatus. Using a suitable dropper, such as a Pasteur pipet, place approximately 0.150 gm (about four drops) of distilled liquid mustard (HD) of at least 95-percent purity on each specimen, inside the attached ring. Cover each specimen with a glass cover plate (see figure 6). Open the drawer, place the glass plate on the drawer support and push in the drawer. Observe the indicator paper by means of the mirror inside the apparatus; a portable light may used to illuminate the indicator paper. The end of the test occurs when the first visible blue color appears where the paper was spotted. If the blue color first appears at the outer edge of the spots and moves inward, suspect a false break from vapors inside the apparatus instead of from agent permeation through the specimen. The time interval between the instant the mustard is placed on the specimen and the first visible appearance of blue on the spotted portion of the indicator paper is termed the mustard resistance of the material.

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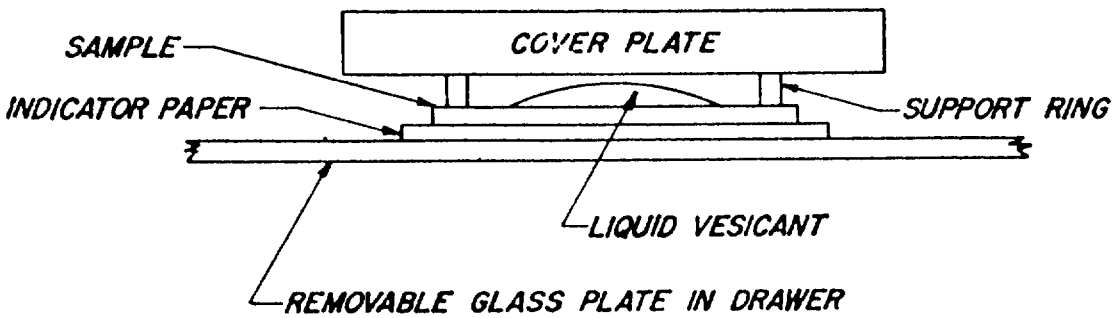


Figure 6. Specimen arrangement, static–diffusion test

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A.6.3 Decontamination. After the test is completed, clean the equipment for reuse by following standing operating procedures that are in effect at the facility performing the testing. It is necessary that all items contaminated with mustard be thoroughly decontaminated, and the reusable items, such as glass plates and cover glasses, be decontaminated and cleaned to the extent that there is no residual mustard that might affect personnel or the indicator paper used in subsequent testing.

A.7 QUALITY CONTROL

A.7.1 Standard reference material.

A.7.1.1 For each type of material test using this method, a standard reference material (SRM) should be made available for the laboratory to use a means of quality control of within day and between day variation. The SRM should simulate the sample material to be tested as closely as possible with respect to formulation and ideally break after the sample material. A sufficient volume of any lot of an SRM should be available to allow the laboratory to test SRM samples on successive days to establish a data base, a statistical quality control chart with control limits and, thereafter, to test the SRM during each test of sample materials for a period of at least six months. A minimum of two SRM samples should be tested each time and results plotted on control charts with established quality control limits. If these limits are breached, sample data should be considered invalid and the test repeated after the cause of excess variation is identified and corrected; a contingency plan for such situations should be documented and made available for laboratory personnel conducting the test. These procedures apply only to those materials for which an SRM is available.

A.7.1.2 SRMs should also be used for proficiency testing of contract laboratories during their pre-certification process and for continuing certification to conduct this test.

A.7.2 Blank. In addition to the SRM, a blank must be tested with each sample batch, the blank being a sample carried through the entire test without mustard being added. The purpose of the blank is to determine if any chemicals emanate from the sample material during the test and cause a positive reaction with the S-328, give the blue color and, therefore, act as a false positive.

A.8 NOTES

A.8.1 Mustard produces severe blister action in both the liquid and vapor phases. All mustard containers should be kept in a well ventilated hood when not in use. Great care should be exercised to prevent either the liquid mustard or its vapor from coming in contact with the body.

A.8.2 Personnel should wear butyl rubber gloves in all operations involving the handling of mustard. The glove protocol in effect for the standing operating procedures being used should be strictly observed.

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A.8.3 If the material tested contains a compound that reacts with S-328, a false end point is obtained. Protective dubbing is an example of such a material. Thus, leather impregnites, such as dubbing, cannot be tested for mustard resistance using S-328-spotted congo red paper and watching for the blue spots to appear. The mustard resistance of materials such as dubbing impregnated leather may be measured by using DB-3 indicator paper in place of the S-328-spotted congo red paper.

A.8.4 If the services of a contracted commercial laboratory are to be procured for performance of this test method, certification of such laboratory must be initiated and maintained by the government during the contractual period. As a source of technical expertise regarding this test method, the government certifying body is to utilize the services of the government laboratory responsible for the routine performance of this test method for production acceptance testing purposes. The government laboratory is to provide the contract laboratory the current standing operating procedures for conduct of this test method, necessary apparatus, SRM, if available, and supplies of necessary reagent solutions and materials. If SRMs are available, the government laboratory is to test and evaluate the proficiency of the contract laboratory to conduct this test method before contract work starts and periodically during the contract. Contract work is not to be performed unless the contractor can demonstrate quality control in the laboratory using SRMs and proficiency in the performance of this test method.

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APPENDIX B

SOLAR RADIATION (SUNSHINE)

B.1 SCOPE

B.1.1 Scope. This appendix describes a test method to determine the effects of solar radiation on the canister. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

B.3 APPARATUS

B.3.1 Test facility. The facility required for this test consists of a chamber or cabinet, auxiliary instrumentation, and a solar lamp bank. This apparatus shall be capable of maintaining and monitoring required conditions of temperature, airflow and irradiation.

B.3.1.1 Chamber. The air velocity in the chamber shall be maintained between 0.25 and 1.5 meters per second (50 to 300 ft/min). The volume of the test chamber shall be a minimum of 10 times that of the envelope volume of the canister.

B.3.1.2 Lamp bank.

B.3.1.2.1 The solar radiation source area shall be such that the length and diameter of the canister shall be no more than one-half the same dimensions of the lamp bank and shall be composed of radiant heat-producing lamps.

B.3.1.2.2 The irradiance shall have a maximum intensity of 1120 Watts per square meter ($\pm$ 10 percent) and the radiation on the test item shall be uniform to within $\pm$ 10 percent of the desired value, with the spectral distribution given in table VIII.

B.3.1.2.3 The radiation source shall be located at least 76 centimeters (30 inches) away from any other surface of the canister.

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Table VIII Spectral energy distribution.

Characteristic	SPECTRAL REGION			
	Ultraviolet		**Visible**	**Infrared**
Bandwith	0.28 to 0.32 µm	0.32 to 0.40 µm	0.40 to 0.78 µm	0.78 to 3.00 µm
Irradiance	5 W/m^2	63 W/m^2	517 to 604 W/m^2	492 W/m^2
Tolerance	±35%	±25%	±10%	±20%

*NOTE: The amount of radiation wavelength shorter than 0.30 µm reaching the Earth's surface is insignificant.

B.3.1.2.4 Light source. The light source used shall satisfy the full spectrum of table VIII and may be one of the following:

(a) Xenon arc or mercury xenon arc (used singularly) with suitable reflector.
(b) Combination of high pressure sodium vapor and improved mercury vapor with suitable reflectors.
(c) High-intensity multivapor, mercury vapor (with suitable reflectors), and incandescent spot lamps.
(d) Carbon arc lamps with suitable reflectors.

Note: Other combinations of the lamps listed above may be used if it is proven that the combination produces the spectrum of table VIII.

B.4 PROCEDURE

B.4.1 The canister shall be exposed to 3 continuous, 24 hour cycles of controlled simulated solar radiation and dry bulb temperatures as specified in table IX.

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Table IX. Solar radiation and dry bulb temperatures

Time	Temperature (°F)	Watts per square meter
0000	98	0
0300	93	0
0600	90	55
0900	101	730
1200	112	1120
1500	119	915
1600	120	730
1800	107	270
2100	105	0
2400	98	0

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